



LOB 1003 - Cálculo 1

Lista de exercícios 3 - Parte 3 2º semestre de 2025

1. 16 pol.
2. 80000 m²
3. 9 pol $\times$ 18 pol
4. $\theta = \frac{\pi}{2}$
5. $r \approx 8,165$ cm, $h \approx 11,547$ cm, $V \approx 2418$ cm³
6. $r = \sqrt{2}$, $h = 1$, $V = \pi \frac{\sqrt{2}}{3}$
7.
 - a) $v = 96$ pés/s
 - b) $t = 3$ s, $s = 256$ pés
 - c) $v = -128$ pés/s
8. Lucro = $nc - nx$, $x = \frac{100 + c}{2}$
9. $x = \sqrt[3]{2}$, $h = \frac{\sqrt[3]{2}}{2}$
10. $r = \frac{\sqrt[3]{15}}{2}$ m, $d = 2\sqrt[3]{15}$ m
11. $V = \frac{32\pi a^3}{81}$
12. (1,2).
14. $\Delta y = \frac{9}{4}$
15. $N = 1$
16. $x = 150$ m e $y = 100$ m

17. 4000 cm^3

18. $\left(-\frac{6}{5}, \frac{3}{5}\right)$

19. $\left(-\frac{1}{3}, \pm \frac{4\sqrt{2}}{3}\right)$

20. $P = \frac{E^2}{4r}$